DATA ENGINE BUILD PLAN · COMPANION TO THE LEAD ENGINE ICP

Legal Soft Scraper Engine

The road to 10,000 quality leads runs through roughly **320,000 raw records**. Three stages destroy 80% of it. This plan names those three, tells you the floor at each one, and stops the build when a floor is missed — because a bad lead is not neutral, it burns a sending domain.

Target 10,000 Grade-A leads **Build** 9 weeks **One-time** $4.2k–6.7k **Run rate** ~$850/mo

§1 Define "quality" before you scrape anything

"10,000 leads" is not a target until "lead" is a testable predicate. Everything downstream — the yield floors, the cost model, the go/no-go gates — is measured against this definition, so it gets written first and does not move.

A record is a **Grade-A lead** only when every one of these is true. Any single failure demotes it; it does not get shipped with a caveat.

GRADE-A PREDICATE · ALL NINE REQUIRED

#	CONDITION	HOW IT IS CHECKED
01	**US private law firm, in practice today**	Website responded 200 within the last 30 days; not a directory stub
02	**1–25 attorneys**	Counted from the team page; confidence flag stored alongside
03	**Practice area in Tier 1A / 1B / 2 / 3**	Classified from site copy, not from a directory category
04	**Fee model is contingency, flat, or fee-shifting**	Detected from fee language on the site — the ICP axis
05	**Named decision-maker with a real title**	Owner / managing partner / firm administrator. Not "info@"
06	**Email verified deliverable**	SMTP-verified *valid*. Catch-all and "risky" both fail
07	**Phone number present**	E.164 normalised, main line minimum
08	**Not on any suppression list**	Existing customer, prior opt-out, competitor account, disqualified size
09	**Enrichment fresher than 90 days**	Timestamped per field, not per record

> GRADE B IS A PRODUCT, NOT A REJECT PILE
>
> Roughly 4,000 records will pass every test except email verification — small firms where no personal mailbox resolves. Those are not waste. They have a named partner, a main line and a live signal, which makes them a **phone-first and LinkedIn-first list**, worked by a different motion.
>
> Ship them as Grade B with the failing field named. Deleting them throws away a quarter of the work; emailing them anyway burns the domain.

§2 The yield model — the whole journey on one page

Read the **yield** column, not the count column. It shows where the funnel actually collapses, and the **floor** column is a stop condition: if a stage comes in under its floor, the pipeline is broken upstream and running the next stage only multiplies the error.

RAW RECORDS TO GRADE-A LEADS · YIELD OF PREVIOUS STAGE · FLOORS ARE STOP CONDITIONS

#	STAGE	RECORDS	YIELD OF PREVIOUS	FLOOR	IF BELOW FLOOR
01	**Raw place records harvested** ~18,000 grid queries: 15 practice keywords × ~1,200 US towns over 15k population	320,000	—	—	—
02	**After identity resolution** COLLAPSE 1 Same firm appears in Places, bar directory and Justia under three names	128,000	40%	30–50%	**Above 50%: dedupe is broken.** You will email the same firm four times
03	**Has a resolvable own website**	95,000	74%	70%	Places field mask is wrong — `websiteUri` not requested
04	**Site live and crawlable**	88,000	93%	85%	Crawler is being blocked — check UA, rate, robots handling
05	**Practice area in ICP tiers** COLLAPSE 2	48,000	55%	45%	**Your keyword grid pulled the wrong firms.** Fix the grid, re-harvest
06	**1–25 attorneys**	44,000	92%	85%	Attorney counter is over-counting staff bios as attorneys
07	**Fee model qualifies**	31,000	70%	60%	Fee-language patterns too narrow — widen before re-running
08	**Decision-maker identified**	24,000	77%	70%	Add the state bar directory join — it names partners for free
09	**Email verified deliverable** COLLAPSE 3 The stage that decides whether the whole engine works	15,500	65%	55%	**Wrong enrichment vendor for SMB legal.** Test a second before scaling
10	**Passes suppression**	14,200	92%	85%	Suppression list is stale or over-broad
11	**All nine fields complete and fresh** GRADE A	11,500	81%	75%	Field-level freshness gaps — refresh, don't re-harvest

11,500 against a 10,000 target is a 15% buffer, which is the right amount — every stage above is an estimate, and a plan that lands at exactly 10,000 has no room to be wrong once. Alongside it, roughly **4,000 Grade-B** records (phone and LinkedIn, no verified email).

> WHAT THE FINISHED ASSET ACTUALLY PRODUCES
>
> 10,000 verified firms is the *standing universe*, not a campaign list. Layered on top, the signal collectors in §8 light roughly **1,200 of them per month** with a live buying signal on a rolling basis — which is the number the Lead Engine funnel math runs on.
>
> The universe is the durable asset. The signals are the revenue.

§3 Tech stack

Every pick below is justified against the specific shape of this workload: ~130,000 domains crawled once, then a small hot subset swept nightly, with fuzzy identity resolution across sources.

BULK CRAWL

`Scrapy 2.12`

The universe crawl — 130k domains, ~4 pages each. Autothrottle, robots.txt, request dedupe, retry middleware and resumable job state are all built in.

Hand-rolling this over `asyncio` means re-implementing throttling and resumability badly. A crawl that cannot resume will be restarted from zero at least twice.

rejected: bare httpx loop, Colly

API CALLS

`httpx` + `tenacity`

Places, DataForSEO, enrichment, verification. Async with bounded concurrency and exponential backoff on 429.

Different failure profile from crawling — paid, rate-limited, and each retry costs money. Keep it out of Scrapy so a crawl retry storm never bills you.

HTML → TEXT

`selectolax` + `trafilatura`

selectolax parses; trafilatura strips nav, footer, cookie banners and ad boilerplate down to main content.

trafilatura cuts LLM input tokens by roughly 60%. On 48,000 extractions that is the difference between a $700 bill and a $280 one, and it improves accuracy by removing the footer that lists every practice area the firm has ever mentioned.

rejected: BeautifulSoup — 3–5× slower at this volume

JS RENDERING

`Playwright`, gated

Only fires when the static fetch returns under 500 words. Expect 5–8% of law firm sites.

Rendering everything costs 20× the compute for a single-digit share of pages. The gate is one `if`, and it is the difference between a two-day crawl and a three-week one.

DATABASE

`Postgres 16` on Cloud SQL

`pg_trgm` for fuzzy name matching, `postgis` for the geo grid, `JSONB` for raw payloads, partial indexes on the hot filters.

Not Firestore. Nightly rescoring reads all 130,000 records — that is 130k billed document reads every night, forever. Dedupe needs trigram similarity joins, which Firestore cannot express at all. This workload is relational and analytical; use the relational, analytical database.

rejected: Firestore, MongoDB

RAW STORAGE

`Cloud Storage`

Gzipped HTML snapshot per fetch, keyed by content hash. Lifecycle rule to Nearline at 30 days.

When the extraction schema changes in week 7 — and it will — you re-run extraction over stored HTML for pennies instead of re-crawling 130,000 domains for a week.

QUEUE / FRONTIER

`Redis`

Crawl frontier, per-domain rate-limit tokens, in-flight dedupe set.

Per-`domain` politeness, not global. One firm's site should never see more than one request every two seconds regardless of how many workers are running.

ORCHESTRATION

`Prefect 3` on Cloud Run Jobs

Python-native flows, typed retries, per-task observability, backfills.

Airflow is heavier than this needs; Temporal is a bigger commitment than a data pipeline warrants. If you want zero new dependencies, Cloud Run Jobs + Cloud Scheduler + Pub/Sub is a legitimate substitute — you trade observability for one less system.

EXTRACTION

`claude-haiku-4-5` via Batch API

Structured outputs (`output_config.format`) for schema-valid JSON. Batch API halves the price. Escalates to `claude-opus-5` on low confidence.

This is bulk extraction over trimmed text — a Haiku-shaped job. Batch is the right mode because nothing here is latency-sensitive; the whole run can take six hours. Full cascade in §7.

PROXIES

Datacenter for firm sites, residential for hard sources

Bright Data, Oxylabs or Decodo.

Law firm sites are overwhelmingly WordPress with no bot protection — polite rate limiting and an honest user agent get through. Save residential IPs for the small number of sources that need them, and the proxy bill drops from four figures to three.

VERIFICATION

`MillionVerifier` or `ZeroBounce`

Every address, before it is ever loaded into a sequencer. Reject catch-all and risky, not just hard-invalid.

Under a tenth of a cent per address. The alternative is a 15% bounce rate in week one, which ends the programme — see §13.

SENDING

`Smartlead` or `Instantly`

Across secondary domains and warmed Google Workspace mailboxes. Never the primary domain.

Full deliverability build in §11. This is infrastructure, not a tool choice — it needs four weeks of lead time and must start in week 1, in parallel with the crawl.

---

§4 Source ledger

Each source is rated for what it uniquely provides, not for how much data it holds. Sources that only duplicate what a cheaper source already gave you are omitted.

DATA SOURCES · WHAT EACH ONE IS THE BEST SOURCE FOR

SOURCE	UNIQUELY PROVIDES	STANDING	BUILD COST
Google Places API Text Search + Place Details	The geographic spine. Name, address, phone, website, category, rating and review count for essentially every firm with a physical office.	OFFICIAL Paid. Note the caching restrictions — store `place_id` and your own derived fields, refresh rather than hoard.	$2,500–3,500
State bar directories All 50 states	**Attorney names, admission dates, firm affiliation, status.** The single best free source for the decision-maker's actual name — and the one generic B2B databases miss entirely.	PUBLIC RECORD Crawl politely; several states publish bulk downloads on request.	$0
The firm's own website	Practice areas, fee model, attorney count, languages, CMS fingerprint, careers page. Every ICP field that matters is here and nowhere else.	PUBLIC robots.txt honoured, rate limited, honest UA.	proxy only
Justia / Avvo / Martindale	Coverage backfill for firms with weak Places presence, plus practice-area cross-check.	CHECK TOS Use as a gap-filler, not a primary spine.	proxy only
DataForSEO Reviews + SERP + Ads	Full Google review history with text and dates — the intake-failure signal. Places Details alone returns only a handful of reviews.	LICENSED Account already exists.	$150–400
Firm careers page JobPosting JSON-LD	The strongest buying signal, from the firm's own server, with a real `datePosted`.	PUBLIC The defensible way to get job signal.	included
Meta Ad Library API	Active ad creative and spend bands — the paid-acquisition signal and mass tort campaign detection.	OFFICIAL FREE API Requires an app and review.	$0
DIY tech fingerprinting	Filevine, Clio, CASEpeer, MyCase, Litify detection — from script tags, chat widgets and form endpoints on the homepage you already fetched.	DERIVED Free; no vendor needed.	$0
Apollo / Findymail / Prospeo	Work email and direct dial for the named partner.	LICENSED Test two on the same 500-record sample before committing.	$600–1,500
Indeed / ZipRecruiter direct	—	DO NOT SCRAPE Against ToS. Use careers pages, Google Jobs via DataForSEO, or a licensed feed instead — you lose almost nothing.	–
LinkedIn	—	DO NOT SCRAPE Account bans plus real legal exposure. Sales Navigator used manually is fine; automated extraction at scale is not.	–

§5 Schema

Six tables. The important design decision is that **every extracted field carries its own source and timestamp** — a record is never uniformly "fresh", and treating it that way is what makes a 90-day freshness rule enforceable.

```
-- the resolved entity. one row per real-world firm.
firm (
  id, canonical_name, domain, phone_e164, addr_hash, lat, lng, state, city,
  attorney_count, attorney_count_conf,      -- 0..1, drives Opus escalation
  practice_areas text[], primary_practice, tier,
  fee_model,                                -- contingency | flat | fee_shifting | hourly | unknown
  spanish_intake bool, cms_detected,
  fit_score int, intent_score int, band,
  status,                                   -- active | suppressed | disqualified
  first_seen, last_crawled
)

-- every raw observation, before resolution. never mutated.
observation (
  id, source, source_key, firm_id, raw jsonb, fetched_at
)

-- the merge audit trail. lets you unpick a bad dedupe.
firm_alias (
  firm_id, alias_name, alias_domain, alias_phone, source, merged_at, merge_rule
)

-- people. bar_number is the join key generic tools do not have.
contact (
  id, firm_id, full_name, title, role,      -- owner | admin | attorney
  email, email_status, email_verified_at,   -- valid | catch_all | risky | invalid
  phone_direct, linkedin_url, bar_number, source, enriched_at
)

-- append-only. intent score is derived from this, never stored raw.
signal (
  id, firm_id, kind, weight, evidence jsonb, -- evidence = the quote for the first line
  observed_at, expires_at
)

-- checked before every send. domain OR email OR firm_id.
suppression (
  scope, value, reason, added_at            -- customer | optout | competitor | dq
)
```

TWO FIELDS THAT PAY FOR THEMSELVES

`signal.evidence` holds the actual review sentence or job-posting title. Without it, the sequencer cannot write the first line, and the first line is the entire difference between a 3% and an 11% reply rate in the Lead Engine funnel.

`attorney_count_conf` is what routes a record to Opus instead of Haiku. One float column replaces a manual QA queue.

§6 Identity resolution — where the project actually dies

Collapse 1 destroys 60% of the records and it is the stage most teams get wrong, because a broken dedupe looks like success: more rows, more leads, more coverage. You find out in week ten, when the same managing partner has received four emails from four of your domains.

Law firms are unusually hard to deduplicate. One firm is legitimately *"Smith & Associates"*, *"Law Offices of John Smith"*, *"Smith Law Firm, P.C."* and *"John Smith, Attorney at Law"* across four sources — and two genuinely different firms in the same city are often *"Smith Law"* and *"Smith Legal"*.

Match rules, strongest first

APPLIED IN ORDER · A MATCH AT ANY LEVEL MERGES THE CLUSTER

#	RULE	KEY	CONFIDENCE
01	**Registered domain**	eTLD+1, lowercased, `www` stripped. Shared-host and directory domains excluded by blocklist.	DECISIVE
02	**Phone**	E.164 normalised. Excludes tracking numbers — if one number maps to 4+ candidate firms, drop it as a shared answering service.	DECISIVE
03	**Address + name stem**	USPS-normalised street + ZIP+4 hash, *and* trigram similarity > 0.55 on the stemmed name. Address alone is not enough — office buildings hold twenty firms.	STRONG
04	**Name similarity within ZIP**	`pg_trgm` similarity > 0.82 on the name with *law, legal, offices, attorneys, PC, LLP, PLLC, &, and* stripped.	REVIEW QUEUE

Run rules 1–3 as connected components (union-find) over the observation set. Rule 4 does **not** auto-merge — it writes to a review queue, sampled by a human. Auto-merging on name similarity alone is how two real firms become one and a genuine prospect silently disappears.

> **THE TEST THAT CATCHES A BROKEN DEDUPE**
> After resolution, query for firms sharing a phone number or a domain. A correct run returns almost nothing. If it returns thousands, rules 1 and 2 did not fire — usually because phones were stored formatted rather than E.164, or because `www.` and trailing slashes were not stripped before hashing. **Run this query before stage 3, not after stage 11.**

§7 Extraction cascade

Three tiers, cheapest first. Each tier only sees what the tier above could not resolve, which is what keeps the total LLM bill under $400 for 48,000 firms.

EXTRACTION TIERS · COST IS FOR THE FULL 48,000-FIRM RUN

TIER	METHOD	RESOLVES	SHARE	COST
1 FREE	Deterministic. `schema.org` JSON-LD (`LegalService` , `Attorney` , `JobPosting`), plus regex for fee language — *"no fee unless we win"*, *"flat fee"*, *"hourly rate"*, *"Hablamos Español"* — and CMS script fingerprints.	Fee model, languages, CMS, job postings, and practice area on well-marked-up sites.	~55%	$0
2 BULK	`claude-haiku-4-5` over trafilatura-trimmed text from homepage, about, team and practice pages. **Batch API** (50% off) with structured outputs for schema-valid JSON.	Practice areas, attorney count, fee model, decision-maker name and title — everything the markup did not declare.	~42%	~$140
3 ESCALATION	`claude-opus-5` , live not batched, on records where tier 2 returned `attorney_count_conf < 0.6` or a fee model of `unknown` — and on every Tier-1A firm regardless of confidence.	The ambiguous cases, plus the highest-value segment where a wrong tier costs a real deal.	~3%	~$120

Total: roughly $260 for the full run, against $4,200–6,700 for the build overall. The extraction that everyone assumes will be the expensive part is under 5% of the cost — because trafilatura removes the boilerplate before the model ever sees it, and Batch halves what remains.

> **TWO THINGS TO GET RIGHT IN THE EXTRACTION PROMPT**
>
> Make the model return `unknown` as a first-class value with a confidence float. A model forced to choose between contingency and hourly will choose, confidently and wrongly, and the ICP axis from the Lead Engine is exactly the field you cannot afford to guess.
>
> Ask for the *evidence span* alongside each field — the sentence it read the fee model from. It makes QA a sampling job instead of a re-reading job, and it is what fills `signal.evidence` .

§8 Signal collectors

The universe crawl runs once. These run forever, and they are what turn 10,000 static records into ~1,200 in-market accounts a month.

ONGOING SWEEPS · CADENCE, SCOPE, COST

COLLECTOR	CADENCE	SCOPE	METHOD	MONTHLY
Job postings TIER 1	Nightly	All 14,000 qualified firms	Fetch `/careers`, `/jobs`, `/employment`; parse `JobPosting` JSON-LD for `title` + `datePosted`. Backfill via Google Jobs through DataForSEO where the site has no careers page. **Track first-seen date yourself** — that is what makes the "open >21 days" signal possible.	$80
Reviews	Weekly	Tier 1A + 1B only (~6,000)	DataForSEO Google Reviews. Filter last 90 days, match against the unreturned-call phrase set, store the matching sentence as evidence.	$60
Paid acquisition	Monthly	Tier 1A (~4,000)	Meta Ad Library API for active creative; Google Ads Transparency via DataForSEO. Mass tort detection = tort keywords in ad copy.	$40
Tech stack change	Quarterly	All qualified	Re-fingerprint the homepage. A CMS that appears where none was = a fresh implementation = signal 09.	$15
Liveness & freshness	Rolling 90-day	All 14,000	HEAD request; on change, re-crawl and re-extract. Keeps the freshness clause of the Grade-A predicate true.	$25

> **THE NIGHTLY JOB SWEEP IS THE HIGHEST-VALUE JOB IN THE SYSTEM**
>
> It is also the one with a hard latency requirement. Signal 01 decays — a posting found on day 2 converts far better than the same posting found on day 20. Run it at a fixed hour, alert on failure, and treat a missed night as an incident, not a retry.

§9 Nine-week build

Two tracks in parallel. The data track builds the universe; the delivery track warms sending infrastructure, which needs four weeks of calendar time and cannot be compressed at the end.

BUILD SEQUENCE · GATE = WHAT MUST BE TRUE TO PROCEED

WK	DATA TRACK	DELIVERY TRACK	GATE
1	Postgres schema, Prefect scaffold, Redis frontier. Build the grid: 15 practice keywords × ~1,200 towns.	Register 5 sending domains. SPF, DKIM, DMARC. 18 Google Workspace mailboxes created.	One grid cell end-to-end into `observation`
2	**Pilot on 3 metros only** — Los Angeles, Houston, Atlanta. Full pipeline, small scope.	Warmup begins on all 18 mailboxes. Do not touch until week 6.	Pilot yields land within §2 floors
3	Full Places harvest. Identity resolution over the whole set. **Run the shared-phone/shared-domain test.**	Warmup continues.	HARD GATE Collapse-1 yield 30–50%
4	Universe crawl: 130k domains, ~4 pages each, HTML to Cloud Storage.	Warmup continues. Draft the five sequences from Lead Engine §9.	Crawl success ≥85%, no source blocking
5	Extraction cascade, all three tiers. Classification and fee-model detection.	Warmup continues.	HARD GATE Collapse-2 yield ≥45%
6	State bar directory join for partner names. Contact enrichment on the qualified set.	Warmup complete. Send volume test at 10/day/mailbox.	Decision-maker found on ≥70%
7	Email verification. Suppression build. **Human QA on a random 200 — eight fields each.**	Sequencer wired to the database. Live-fire on 50 leads.	HARD GATE QA accuracy ≥95%, bounce <2%
8	Signal collectors live. Fit and intent scoring. Nightly rescore job.	Scale to 40/day/mailbox. Reply handling and routing.	Signals firing on ≥1,000 accounts
9	**10,000 Grade-A + ~4,000 Grade-B delivered.** Refresh loop running.	Full cadence. Attribution back to originating signal.	Signal-to-contact latency under 48h

> WEEK 2 IS NOT OPTIONAL
>
> Three metros, full pipeline, every stage measured. It costs four days and it is the only cheap opportunity to discover that the keyword grid is pulling estate-planning firms, or that the fee-language regex misses *"contingency"* spelled as *"contingent fee basis"*.
>
> Finding that in week 5, after the full harvest, means re-running Places at full cost. The pilot is the single highest-return four days in the plan.

§10 Cost model

ONE-TIME BUILD

ITEM	COST
Google Places API ~18k Text Search + ~130k Details	$2,500–3,500
Contact enrichment ~30k lookups	$600–1,500
Proxies	$200–400
DataForSEO reviews + SERP backfill	$150–400
LLM extraction Haiku batch + Opus escalation	$260–400
Cloud SQL build months	$300
Cloud Run + storage	$150–300
Email verification ~24k	$30–100
Total	**$4,190–6,900**

MONTHLY RUN RATE

ITEM	COST
Cloud SQL Postgres	$120
Cloud Run + storage	$60
Signal collectors §8 total	$220
Enrichment refresh new signal accounts	$150
Sending 18 mailboxes + sequencer + domains	$250
LLM new firms + reclassification	$50
Total	**~$850**

Excludes SDR salaries, which sit in the Lead Engine funnel model, not here.

WHAT A LEAD ACTUALLY COSTS

Take the midpoint: $5,500 to produce 10,000 Grade-A plus 4,000 Grade-B records. That is $0.39 **per qualified, verified, decision-maker-attached lead** — against an account ACV of $35,900 and a published allowable CAC of $2,500.

The data layer is not where this business is cost-constrained. Which is the argument for spending properly on it — better verification, a second enrichment vendor, the Opus escalation tier — rather than saving $200 and shipping a list that bounces.

§11 Delivery infrastructure

10,000 perfect leads produce nothing if the mail lands in spam. This track starts in week 1 and runs in parallel, because warmup is calendar time that cannot be bought back.

ELEMENT	SPEC	WHY
Sending domains	5 secondary domains. Never `legalsoft.com`.	The primary domain carries client mail, invoices and the app. A cold-outbound reputation hit there is unrecoverable and affects the whole business.
Mailboxes	18 — 3–4 per domain, Google Workspace.	Real mailboxes on a real provider deliver better than SMTP relays. Spreading across domains means one flagged domain is not one flagged programme.
Authentication	SPF, DKIM and DMARC on every domain before the first send.	Missing DMARC alone is enough for bulk filtering at Google and Microsoft.
Warmup	4 weeks minimum, automated, ramping to 40/day.	A new domain sending 40 cold emails on day one is a spam trap by definition.
Steady volume	40/day/mailbox → 720/day → ~15,000/month.	10,000 leads × 4 touches = 40,000 sends, worked through in roughly ten weeks.
Bounce ceiling	Hard stop at 2%. Pause the domain, do not push through.	Above 3% the domain is effectively finished. This is the number that verification in §1 exists to protect.
CAN-SPAM	Physical postal address in every email, a working opt-out honoured within 10 days, no deceptive headers or subject lines.	B2B cold email is lawful in the US on these conditions. The Woodland Hills address satisfies the first.
Global suppression	Every opt-out writes to `suppression` at *domain* scope, checked before every send across all sequences.	One partner opting out must silence the whole firm, not just that mailbox.

§12 Legal ledger

Stated plainly so the boundaries are a decision rather than an accident. This is an engineering summary, not legal advice — have counsel confirm before go-live.

ACTIVITY	STANDING	CONDITION
Crawling firm websites	CLEAR	Public pages, robots.txt honoured, rate limited, identifiable user agent, no login bypass.
State bar directories	CLEAR	Public records. Respect per-site terms and rate limits; ask for a bulk file where offered.
Google Places API	CLEAR	Licensed use. Observe the caching and display restrictions — keep `place_id` and your own derived fields, refresh rather than warehouse.
Meta Ad Library API	CLEAR	Official API, app review required.
DataForSEO	CLEAR	Licensed provider, existing account.
Job boards, direct scrape	AVOID	Against ToS. Careers pages plus Google Jobs plus a licensed feed cover the same signal.
LinkedIn, automated	AVOID	Bans plus real legal exposure. Manual Sales Navigator use is a different thing.
B2B cold email	LAWFUL	CAN-SPAM conditions in §11. Suppress on request, immediately and globally.
Cold calling	CONDITIONAL	Business lines are outside most DNC scope, but scrub against the National DNC anyway — solos often list a mobile as the firm number. State laws vary.

§13 Failure modes

Ranked by how often they end the programme rather than merely delay it.

#	FAILURE	HOW IT SHOWS UP	PREVENTION
01	**Skipping email verification** FATAL	15% bounce in week one. Domains blacklisted before a single reply. Ten weeks of work unusable.	Verify every address. Reject catch-all and risky, not just hard-invalid. It costs under a tenth of a cent.
02	**Broken identity resolution** FATAL	Looks like success — more rows than expected. Surfaces as a managing partner receiving four emails from four domains and reporting all of them.	Collapse-1 floor as a hard gate in week 3. Run the shared-phone/shared-domain query before proceeding.
03	**Sending from the primary domain** FATAL	Client mail and app notifications start landing in spam. Not recoverable on any useful timescale.	Five secondary domains, provisioned week 1. This is not a shortcut worth taking once.
04	**Full harvest before the pilot**	Week 5 reveals the grid pulled the wrong practice areas. Places spend already committed.	The three-metro pilot in week 2. Four days against a $3,000 re-run.
05	**Firestore as the lead store**	Nightly rescore bills 130k document reads. Dedupe cannot be expressed at all, so it gets skipped — which causes failure 02.	Postgres. The one architectural decision that is hard to reverse later.
06	**No suppression check**	Cold outbound reaches an existing Legal Soft customer. Worse than a wasted send — it damages a paying account.	Load the customer list into `suppression` at domain scope on day one, before the first send.
07	**Not storing raw HTML**	The extraction schema changes in week 7 and the only way to backfill is re-crawling 130,000 domains.	Cloud Storage, gzipped, content-hash keyed. Cheaper than one re-crawl.
08	**Job sweep silently failing**	No error, just no new signals. Discovered weeks later when pipeline dries up.	Alert on *absence* — if the nightly sweep finds zero new postings across 14,000 firms, that is a failure, not a quiet night.

> WHERE TO START ON MONDAY
>
> Two things, in parallel, before anything else: **register the five sending domains and start warmup** — it is four weeks of calendar time and nothing else in this plan can shorten it — and **stand up Postgres with the §5 schema**, then run the three-metro pilot against it.
>
> Everything else in this document is downstream of those two, and both can be done this week.

Companion to the Legal Soft Lead Engine ICP · tier definitions, signal weights and funnel conversion assumptions are carried from that document.

Yield percentages and record counts are modelled estimates to be replaced with pilot measurements in week 2. Google Places, enrichment and proxy pricing move — verify current SKU rates before committing budget. LLM costs computed at published Claude API rates with Batch API pricing applied to tier 2. The legal ledger is an engineering summary, not legal advice.